1 Solution — GIAM 8.2 Problem 4

1.1 Problem

Determine a formula for the bijection from $(-1, 1)$ to the line $y = 1$ determined by vertical projection onto the upper half of the unit circle, followed by projection from the point $(0, 0)$.

1.2 Setup — Parsing the Two-Step Construction

The construction is a *composition* of two geometric projections:

1. **Vertical projection**: each $x \in (-1, 1)$, regarded as the point $(x, 0)$ on the x-axis, is sent straight up to the unique point on the upper half of the unit circle $x^2 + y^2 = 1$, $y > 0$ directly above it.
2. **Central projection from the origin**: the resulting point on the half-circle is sent along the ray from $(0, 0)$ through it, continuing until that ray hits the horizontal line $y = 1$.

The bijection f is the composite. Our job is to write down $f(x)$ as a formula.

1.3 The Geometric Picture

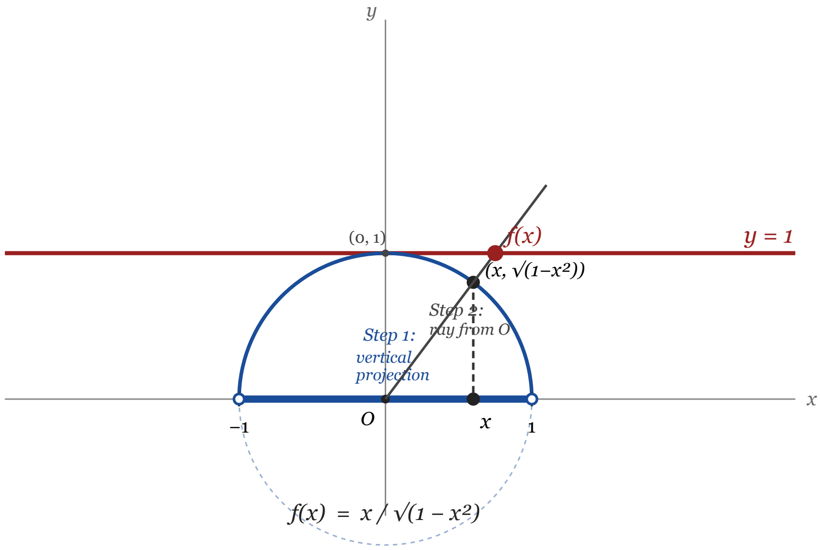

Figure 1.1: The two-step bijection from (−1, 1) to the line y = 1. Step 1 sends x ∈ (−1, 1) straight up to the point (x, √(1−x²)) on the upper unit half-circle. Step 2 follows the ray from the origin through that point until it hits y = 1, landing at the image f(x) = x / √(1 − x²).

1.4 Step 1 — Vertical Projection to the Upper Half-Circle

For $x \in (-1, 1)$, the point on the upper unit circle directly above $(x, 0)$ has y-coordinate $\sqrt{1-x^2}$ (the positive root of $y^2 = 1 - x^2$). Write this map as

$$g_1 : (-1, 1) \longrightarrow S^+, \qquad g_1(x) = \big(x,\ \sqrt{1-x^2}\big),$$

where $S^+ = \{\,(x, y) : x^2 + y^2 = 1,\ y > 0\,\}$ is the open upper half-circle. The map g_1 is a bijection: given any $(a, b) \in S^+$, the unique preimage is a, so $g_1^{-1}(a, b) = a$.

1.5 Step 2 — Central Projection from $(0, 0)$ onto $y = 1$

Given a point $P = (a, b) \in S^+$ (so $b > 0$), the ray from $(0, 0)$ through P is

$$\{\,(ta,\ tb)\ :\ t \geq 0\,\}.$$

It meets the horizontal line $y = 1$ when $tb = 1$, i.e., at $t = 1/b$; the meeting point's x-coordinate is $ta = a/b$. Identifying the line $y = 1$ with $\mathbb{R}$ via the x-coordinate, this gives

$$g_2 : S^+ \longrightarrow \mathbb{R}, \qquad g_2(a,b) \;=\; \frac{a}{b}.$$

g_2 is a bijection: given $u \in \mathbb{R}$, the unique preimage is the point on S^+ along the ray from $(0,0)$ through $(u,1)$, namely $(u,1)/\sqrt{u^2+1} = \big(u/\sqrt{u^2+1},\, 1/\sqrt{u^2+1}\big)$.

1.6 The Composite Formula

Compose the two steps:

$$f(x) \;=\; g_2\big(g_1(x)\big) \;=\; g_2\big(x,\, \sqrt{1-x^2}\big) \;=\; \frac{x}{\sqrt{1-x^2}}.$$

So the desired bijection is

$$\boxed{f \,:\, (-1,1) \longrightarrow \mathbb{R}, \qquad f(x) \;=\; \frac{x}{\sqrt{1-x^2}}}$$

(under the identification of the line $y=1$ with $\mathbb{R}$ by x-coordinate).

1.7 Verifying f is a Bijection

f is the composite of two bijections g_1, g_2, hence is itself a bijection by transitivity of $\equiv$ (Section 8.1, Problem 2). To make this concrete, exhibit the inverse.

1.7.1 The Inverse

Solve $y = x/\sqrt{1-x^2}$ for x. Squaring both sides,

$$y^2 \;=\; \frac{x^2}{1-x^2} \quad\Longrightarrow\quad y^2(1-x^2) \;=\; x^2 \quad\Longrightarrow\quad y^2 \;=\; x^2(1+y^2) \quad\Longrightarrow\quad x^2 \;=\; \frac{y^2}{1+y^2}.$$

Taking the positive root and matching signs (since x and y have the same sign — the construction sends positive x to the right and negative x to the left),

$$f^{-1}(y) \;=\; \frac{y}{\sqrt{1+y^2}}.$$

A quick check: f^{-1} is exactly $g_1^{-1} \circ g_2^{-1}$. Indeed, $g_2^{-1}(y) = (y,1)/\sqrt{1+y^2}$ as computed above; then projecting to the x-axis (taking the first coordinate) gives $y/\sqrt{1+y^2}$.

Note $f^{-1}(y) \in (-1,1)$ for all $y \in \mathbb{R}$: since $|y| < \sqrt{1+y^2}$, we have $|f^{-1}(y)| < 1$. ✓

1.8 Sanity Checks

x	$\sqrt{1-x^2}$	$f(x) = x/\sqrt{1-x^2}$
0	1	0
$1/2$	$\sqrt{3}/2$	$1/\sqrt{3} \approx 0.577$
$3/5$	$4/5$	$3/4 = 0.75$
$4/5$	$3/5$	$4/3 \approx 1.333$
$\to 1^-$	$\to 0^+$	$\to +\infty$
$\to -1^+$	$\to 0^+$	$\to -\infty$

Table 1.1

The $x = 3/5$ row matches the geometric picture above (the highlighted worked example uses $x = 3/5 = 0.6$, landing at $f(x) = 3/4 = 0.75$).

The endpoint behavior gives the right *qualitative* picture: as x slides toward ± 1, the corresponding circle point approaches $(\pm 1, 0)$, the ray from the origin through it tilts toward horizontal, and the meeting with $y = 1$ races off to $\pm\infty$. So f stretches $(-1,1)$ onto the whole of $\mathbb{R}$.

1.9 Remark — A Trigonometric View

Parametrize the upper half-circle by the angle $\theta \in (0,\pi)$ from the positive x-axis: the half-circle point is $(\cos\theta, \sin\theta)$. Then:

- $g_1(x) = (x, \sqrt{1-x^2}) = (\cos\theta, \sin\theta)$ with $\theta = \arccos(x)$.
- $g_2(\cos\theta, \sin\theta) = \cos\theta/\sin\theta = \cot\theta$.

So f is the composition $x \mapsto \arccos(x) \mapsto \cot(\arccos(x))$, i.e.,

$$f(x) \;=\; \cot(\arccos x) \;=\; \frac{\cos(\arccos x)}{\sin(\arccos x)} \;=\; \frac{x}{\sqrt{1-x^2}},$$

recovering the same formula. The vertical projection is the *inverse cosine*, and the central projection from the origin is the *cotangent* — together, $f = \cot \circ \arccos$.

In particular, since $\cot : (0, \pi) \to \mathbb{R}$ is a known bijection (monotonically decreasing from $+\infty$ at 0 to $-\infty$ at π), composing with the bijection $\arccos : (-1, 1) \to (0, \pi)$ confirms f is a bijection directly.

1.10 Remark — Why This Construction Matters

This problem completes a chain begun in Problem 2: any non-degenerate open interval is in bijection with the whole real line. Specifically,

$$(-1, 1) \;\equiv\; \mathbb{R} \quad \text{via } f(x) = \frac{x}{\sqrt{1-x^2}}.$$

Combined with Problem 2 (which gives $(a, b) \equiv (-1, 1)$ via the affine map $x \mapsto -1 + 2(x-a)/(b-a)$) and transitivity of $\equiv$,

$$(a, b) \;\equiv\; (-1, 1) \;\equiv\; \mathbb{R}$$

for every $a < b$. So a one-millimeter open interval has the same cardinality as the entire real line — exactly the §8.2 punch line the textbook foreshadows ("an infinitely extended line has the same number of points as the interval $(0, 1)$").

1.11 Remark — Closed Intervals Need an Extra Step

The bijection f blows up at ± 1, so it does *not* extend to a bijection $[-1, 1] \to (\text{line } y = 1)$: the endpoints ± 1 have no image under f, and conversely the line $y = 1$ has no "point at infinity" for them to map to. To biject a *closed* interval $[-1, 1]$ with $\mathbb{R}$, one needs a more careful construction (e.g., the Cantor–Bernstein–Schröder theorem from §8.4, or an explicit ad-hoc shuffle of countably many points to absorb the two extra endpoints).

The open-interval case in this problem is the clean version: the open ends of $(-1, 1)$ correspond exactly to "approaching the horizontal" — the only way for the central-projection ray to miss the line $y = 1$ — and these directions correspond to $\pm\infty$, which are not points of $\mathbb{R}$. The geometry conspires perfectly with the open-interval setup.

1.12 Remark — Why This Specific Combination of Projections?

Two natural alternatives, for comparison:

- *Stereographic projection from* $(0, 1)$ *onto the* x*-axis.* Each point on the (closed) unit circle except $(0, 1)$ maps to a unique point of the x-axis. Used in complex analysis to add a "point at infinity" to $\mathbb{C} \cup \{\infty\}$.
- *Radial projection from* $(0, 0)$ *onto the unit circle.* Bijection between rays from the origin and points of the circle, but doesn't directly give a bijection to a line.

The textbook's two-step construction (vertical then central) is the natural composite that "unrolls" the bounded open interval $(-1, 1)$ onto the unbounded line. The key insight is that the *bounded* interval $(-1, 1)$ is wrapped around the *bounded* upper half-circle, and then *unbounded* central projection from the origin stretches the half-circle out to the full line. Vertical projection handles the "geometry"; central projection handles the "infinite stretch".